



$$\lim_{x \rightarrow 2} \frac{1}{x-2} \int_{2x}^{x^2} \frac{1}{1+t^3} dt \quad \text{limitinin değeri aşağıdakilerden}$$

hangisidir?

- A) 0 B) 1 C) ∞ **D) $\frac{2}{65}$** E) $\frac{4}{65}$

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{\int_{2x}^{x^2} \frac{dt}{1+t^3}}{x-2} &= \lim_{x \rightarrow 2} \frac{(2x) \cdot \frac{1}{1+(x^2)^3} - 2 \cdot \frac{1}{1+(2x)^3}}{1} \\ &\hookrightarrow 0/0 \rightarrow L'H \\ &= \lim_{x \rightarrow 2} \frac{2x}{1+x^6} - \frac{2}{1+8x^3} = \frac{4}{65} - \frac{2}{65} \\ &= \frac{2}{65} \end{aligned}$$

2

$$h(x) = 8 + \int_1^{x^2} \frac{dt}{\cos^2(t-1)} \text{ fonksiyonunun } x=1 \text{ noktasında}$$

lineer ifadesi aşağıdakilerden hangisidir?

- A) $L(x) = 2x + 6$ B) $L(x) = 2x + 10$ C) $L(x) = 2x + 8$
 D) $L(x) = x + 6$ E) $L(x) = x + 8$

$$L(x) = h(1) + h'(1) \cdot (x-1) \text{ hesaplamalıyız.}$$

$$h(1) = 8 + \int_1^1 \frac{dt}{\cos^2(t-1)} = 8$$

$$h'(x) = 2x \cdot \frac{1}{\cos^2(x^2-1)} \rightarrow h'(1) = \frac{2}{\cos^2 0} = 2$$

$$L(x) = \underbrace{h(1)}_8 + \underbrace{h'(1)}_2 \cdot (x-1) = 8 + 2x - 2 = 2x + 6$$

3) $\int_1^{\ln x} f(t) dt = e^x \sin(\ln x)$ ise $f(0)$ değeri nedir? ($x > 0$)

- A) 0 B) e C) e^2 D) 1 E) $\sin 1$

↓ Türev alalım

$$\frac{1}{x} \cdot f(\ln x) = e^x \cdot \sin(\ln x) + e^x \cdot \frac{1}{x} \cdot \cos(\ln x)$$

↓ $f(0)$ bulmak için $x=1$ koyalım

$$f\left(\frac{\ln 1}{1}\right) = e \cdot \underbrace{\sin\left(\frac{\ln 1}{1}\right)}_0 + e \cdot \underbrace{\cos\left(\frac{\ln 1}{1}\right)}_1 \Rightarrow f(0) = e$$

4) $\lim_{x \rightarrow 0} \left(\frac{1}{x^3} \int_0^x \frac{e^{t^4} - e^{-t^4}}{t^2} dt \right) = ?$ A) $\frac{1}{3}$ B) 1 C) $\frac{2}{3}$ D) $-\frac{1}{3}$ E) $-\frac{2}{3}$

$$\lim_{x \rightarrow 0} \frac{\int_0^x \frac{e^{t^4} - e^{-t^4}}{t^2} dt}{x^3} = \lim_{x \rightarrow 0} \frac{\frac{e^{x^4} - e^{-x^4}}{x^2}}{3x^2} = \lim_{x \rightarrow 0} \frac{e^{x^4} - e^{-x^4}}{3x^4}$$

$\downarrow$
 $0/0 \rightarrow L'H.$

$$= \lim_{x \rightarrow 0} \frac{4x^3 \cdot e^{x^4} + 4x^3 \cdot e^{-x^4}}{12x^3}$$

$$= \lim_{x \rightarrow 0} \frac{4x^3 (e^{x^4} + e^{-x^4})}{12x^3} = \frac{8}{12} = \frac{2}{3}$$

5) $\int (\tan^2 x - \cot^2 x) dx$ integrali aşağıdakilerden hangisidir?

A) $\sec x + \csc x + c$ B) $-\tan x + \cot x - 2x + c$ C) $\tan x - \cot x + 2x + c$

D) $\tan x + \cot x + c$ E) $\cot x - \csc x + c$

1 ekleyip 1 çıkaralım

$$\int (\tan^2 x - \cot^2 x) dx = \int \left(\frac{1 + \tan^2 x}{\sec^2 x} - \frac{1 + \cot^2 x}{\csc^2 x} \right) dx$$

$$= \int (\sec^2 x - \csc^2 x) dx = \tan x + \cot x + c$$

$$\ln \left(\frac{x \cdot (x^3 - 1)}{x^3 - x} \right) = \ln 1 = 0$$

$\uparrow$

6) $\int (1 + \ln x + \ln(x^2 - 1) - \ln(x^3 - x)) dx$ integralinin değeri hangisidir?

A) c B) $\frac{1}{x} + \frac{2x}{x^2 - 1} - \frac{(3x^2 - 1)}{x^3 - x} + c$ C) $x \ln x - x \ln(x^2 - 1) + c$ D) $x + c$

$$\int dx = x + c$$

7) $\int \left(\frac{dx}{4+9x^2} \right)$ integralinin değeri aşağıdakilerden hangisidir?

A) $\frac{1}{3} \ln \left| \frac{9x}{4} \right| + c$

C) $\frac{1}{6} \ln \left| \frac{3x}{2} \right| + c$

E) $\frac{1}{6} \arctan \frac{3x}{2} + c$

B) $\ln|x| + c$

D) $\arcsin \frac{3x}{2} + c$

C) $\frac{1}{6} \ln \left| \frac{3x}{2} \right| + c$

LURAL : $\int \frac{dx}{a^2+x^2} = \frac{1}{a} \arctan \frac{x}{a} + c$

D) $\arcsin \frac{3x}{2} + c$

$$\int \frac{dx}{4+9x^2} = \frac{1}{9} \int \frac{dx}{\left(\frac{2}{3}\right)^2 + x^2} = \frac{1}{9} \cdot \frac{1}{\frac{2}{3}} \arctan \frac{x}{\frac{2}{3}} + c$$

$$= \frac{1}{6} \arctan \frac{3x}{2} + c$$

8) $\int e^{-2\ln(\sqrt{x})} dx = ?$

A) c

B) $\ln x + c$

C) $x + c$

D) $\frac{1}{\sqrt{x}} + c$

E) $\frac{1}{\sqrt[3]{x^2}} + c$

~~*~~ $e^{\ln f(x)} = f(x)$

$$e^{-2\ln(\sqrt{x})} = e^{\ln(\sqrt{x})^{-2}} = e^{\ln(\frac{1}{x})} = \frac{1}{x}$$

$$\int e^{-2\ln \sqrt{x}} dx = \int \frac{dx}{x} = \ln x + c$$

$$\textcircled{g} \int_0^3 g(x) dx = 7, \quad \int_0^6 \underline{g(2x)} dx = 5 \Rightarrow \int_1^2 x \cdot g(3x^2) dx = ?$$

a) 2

b) $\frac{1}{2}$

c) 10

d) 3

e) $\frac{5}{6}$

$$\int_1^2 x \cdot g(3x^2) dx = \frac{1}{6} \int_3^{12} g(u) du = \frac{1}{6} \left[\underbrace{\int_0^{12} g(u) du}_{\textcircled{*} \textcircled{*} \textcircled{*}} - \underbrace{\int_0^3 g(u) du}_{\textcircled{*} \text{ dan } 7} \right]$$

$$3x^2 = u \quad 6x dx = du$$

$$\underline{E.S} \quad \underline{4.S}$$

$$x=1 \rightarrow u=3$$

$$x=2 \rightarrow u=12$$

$$\textcircled{*} \textcircled{*} \text{ dan : } \int_0^6 g(2x) dx = 5 \Rightarrow \frac{1}{2} \int_0^{12} g(u) du = 5$$

$$\downarrow$$

$$2x = u \rightarrow 2dx = du$$

$$\underline{E.S} \quad \underline{4.S}$$

$$x=0 \rightarrow u=0$$

$$x=6 \rightarrow u=12$$

$$\int_0^{12} g(u) du = 10$$

$\textcircled{*} \textcircled{*} \textcircled{*}$ in
sonucu

Sonuç:

$$\int_1^2 x \cdot g(3x^2) dx = \frac{1}{6} \left[\underbrace{\int_0^{12} g(u) du}_{10} - \underbrace{\int_0^3 g(u) du}_7 \right]$$

$$= \frac{1}{6} \cdot 3 = \frac{1}{2}$$

10 $\int \frac{\arctan(\ln x)}{x(1+\ln^2 x)} dx = ?$

A) $\arctan x + c$ B) $\frac{1}{2} \arctan(\ln x) + c$ C) $x \arctan(\ln x) + c$

D) $\frac{1}{2} (\arctan(\ln x))^2 + c$ E) $\arctan(\ln x) + \arctan x + c$

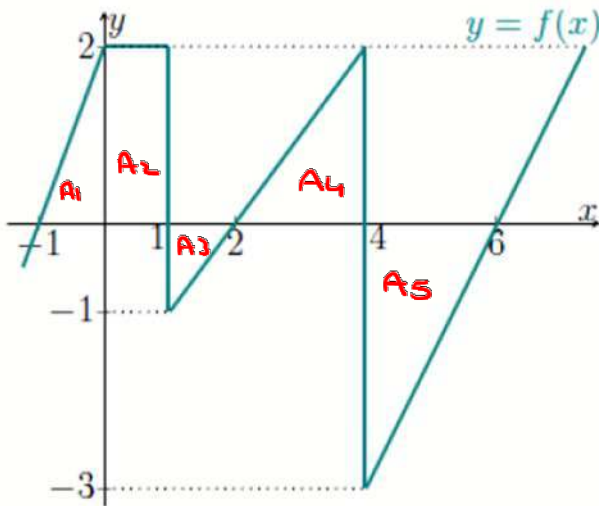
$$\int \frac{\arctan(\ln x)}{x(1+\ln^2 x)} dx = \int \frac{\arctan u}{1+u^2} du = \int t \cdot dt$$

$$\ln x = u \quad \frac{dx}{x} = du \quad \left\{ \begin{array}{l} t = \arctan u \\ dt = \frac{du}{1+u^2} \end{array} \right. \quad \left\{ \begin{array}{l} = \frac{t^2}{2} + c \\ = \frac{(\arctan u)^2}{2} + c \\ = \frac{(\arctan(\ln x))^2}{2} + c \end{array} \right.$$

11 Aşağıda grafiği verilen $y = f(x)$ fonksiyonu için

$$\int_{-1}^{1/2} f(x) dx + \int_3^6 f(x) dx + \int_{1/2}^3 f(x) dx = ?$$

$A_1 = 1$
 $A_2 = 2$
 $A_3 = \frac{1}{2}$
 $A_4 = 2$
 $A_5 = 3$



$$\int_{-1}^{1/2} + \int_{1/2}^3 + \int_3^6 = \int_{-1}^6 f(x) dx$$

soruluyor.

A) $\frac{3}{2}$ B) $\frac{5}{2}$ C) $\frac{9}{2}$ D) $\frac{11}{2}$ E) $\frac{17}{2}$

x-ekseni üstündeki alanların toplamından

x-ekseni altındaki alanları çıkarmalıyız.

$$\int_{-1}^6 f(x) dx = A_1 + A_2 + A_4 - (A_3 + A_5)$$

$$= 1 + 2 + 2 - \left(\frac{1}{2} + 3\right)$$

$$= 5 - \frac{7}{2} = \frac{3}{2}$$

12) f fonksiyonu $[0,4]$ aralığında tanımlı sürekli bir fonksiyon ve ters türevi de F olmak üzere

$$\int_0^4 f(x)dx = 10! \text{ ve } F(0) = 8! \text{ ise } F(4) = ?$$

A) $10! + 8!$

B) $10! - 8!$

C) 0

D) $2!$

E) $4!$

f in ters türevi F ise :

$$10! = \int_0^4 f(x)dx = F(x) \Big|_0^4 = F(4) - F(0)$$

$$10! = F(4) - \underbrace{F(0)}_{8!} \Rightarrow F(4) = 10! + 8!$$

13) $\begin{cases} F(x) = (f \circ g)(x) \\ g(2) = \frac{\pi}{4}; g(-1) = \frac{\pi}{3} \\ f(x) = \tan x \end{cases}$ bilgileri veriliyor. F fonksiyonu h fonksiyonunun bir ters türevi

olduğuna göre $\int_{-1}^2 h(x)dx = ?$

A) $\sqrt{2} - \sqrt{3}$

B) $\sqrt{3}$

C) $1 - \sqrt{3}$

D) 1

E) $\sqrt{3} - \sqrt{2}$

f, h in ters türevi ise ;

$$\int_{-1}^2 h(x)dx = F(x) \Big|_{-1}^2 = F(2) - F(-1)$$

$$= f(\underbrace{g(2)}_{\pi/4}) - f(\underbrace{g(-1)}_{\pi/3})$$

$$= f\left(\frac{\pi}{4}\right) - f\left(\frac{\pi}{3}\right) = \tan \frac{\pi}{4} - \tan \frac{\pi}{3}$$

$$= 1 - \sqrt{3}$$

14) $f(x) = \begin{cases} \sin x \cos x, & 0 \leq x \leq \frac{\pi}{2} \\ \cos^2 x, & \frac{\pi}{2} < x \leq \pi \end{cases}$ fonksiyonu için $\int_0^{\pi} f(x) dx$ belirli integralinin değeri kaçtır?

A) 1

C) $\frac{\pi}{4}$

E) $\frac{\pi+1}{2}$

B) $\frac{\pi}{2}$

D) $\frac{\pi}{4} + \frac{1}{2}$

$$\int_0^{\pi} f(x) dx = \int_0^{\pi/2} f(x) dx + \int_{\pi/2}^{\pi} f(x) dx = \int_0^{\pi/2} \underbrace{\sin x}_{u} \cdot \underbrace{\cos x}_{du} dx + \int_{\pi/2}^{\pi} \underbrace{\cos^2 x}_{\frac{1+\cos 2x}{2}} dx$$

$\frac{u^2}{2} \rightarrow \frac{(\sin^2 x)}{2}$

$$= \underbrace{\frac{\sin^2 x}{2}} \Big|_0^{\pi/2} + \left(\frac{x}{2} + \frac{\sin 2x}{4} \right) \Big|_{\pi/2}^{\pi} =$$

$$= \frac{1}{2} - 0 + \left(\frac{\pi}{2} + 0 - \left(\frac{\pi}{4} - 0 \right) \right) = \frac{1}{2} + \frac{\pi}{4}$$

15) $\int_{-\pi/4}^{\pi/4} [\tan^3(x^3) + \tan^2 x] dx$ belirli integralinin değeri kaçtır?

A) 0

B) $\sqrt{2} - \frac{\pi}{2}$

C) $2 - \frac{\pi}{2}$

D) π

E) $1 + \frac{\pi}{2}$

$$\int_{-\pi/4}^{\pi/4} (\underbrace{(\tan x^3)^3}_{\text{Tek Fonk.}} + \tan^2 x) dx = \int_{-\pi/4}^{\pi/4} \underbrace{\tan^2 x}_{\text{Çift Fonk.}} dx$$

$\left\{ -\frac{\pi}{4}, \frac{\pi}{4} \right\}$ integrali
0 ✓

$$= 2 \int_0^{\pi/4} (\sec^2 x - 1) dx$$

$$= 2 \left[\tan x - x \right]_0^{\pi/4} = 2 \left[1 - \frac{\pi}{4} \right] = 2 - \frac{\pi}{2}$$

16) $\int_{-2}^2 (x^3 \cos^3 x - x - x^5 \sin^2 x + x^2 \sin x + 3) dx = ?$

(A) 12
B) 0
C) 2
D) 4
E) 8

T → Tek Fonk.
G → Çift Fonk.

$$I = \int_{-2}^2 3 dx = 2 \int_0^2 3 dx = 2 \cdot 3 \times \left| \right|_0^2 = 12$$

17) Aşağıda verilen belirsiz integraller ile çözümleri için yapılabilecek dönüşümler hangi seçenekte doğru olarak eşleştirilmiştir?

1. $\int \frac{dx}{2 - \sin x}$

a. $x = \frac{\sqrt{7}}{2} \tan t - \frac{1}{2}$

2. $\int \frac{dx}{\sqrt{x^2 + x + 2}}$

b. $x = \frac{\sqrt{2}}{3} \sin t$

3. $\int \sqrt{2 - 9x^2} dx$

c. $x = 3 \sec t + 1$

4. $\int \frac{dx}{\sqrt{x^2 - 2x - 8}}$

d. $\tan \frac{x}{2} = t$

A) $\begin{matrix} 1 - a \\ 2 - b \\ 3 - c \\ 4 - d \end{matrix}$

(B) $\begin{matrix} 1 - d \\ 2 - a \\ 3 - b \\ 4 - c \end{matrix}$

C) $\begin{matrix} 1 - c \\ 2 - b \\ 3 - d \\ 4 - a \end{matrix}$

(D) $\begin{matrix} 1 - d \\ 2 - a \\ 3 - c \\ 4 - b \end{matrix}$

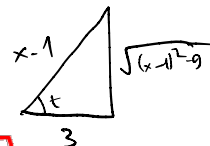
E) $\begin{matrix} 1 - d \\ 2 - c \\ 3 - a \\ 4 - b \end{matrix}$

1. $\int \frac{dx}{2 - \sin x} \rightarrow \sin x, \cos x$ içeren kesirli int. $\Rightarrow x = \tan \frac{x}{2} \rightarrow D$

2. $\int \frac{dx}{\sqrt{x^2 + x + 2}} \rightarrow \sqrt{x^2 + x + 2} = \sqrt{(x + \frac{1}{2})^2 + \frac{7}{4}} \rightarrow x + \frac{1}{2} = \frac{\sqrt{7}}{2} \tan t$
 $x = \frac{\sqrt{7}}{2} \tan t - \frac{1}{2} \rightarrow A$

3. $\int \sqrt{2 - 9x^2} dx$
 $(\sqrt{2})^2 (3x)^2$

$3x = \sqrt{2} \sin t \rightarrow B$



$\cos t = \frac{3}{x-1} \quad \frac{1}{\cos t} = \frac{x-1}{3}$

4. $\int \frac{dx}{\sqrt{x^2 - 2x - 8}} \rightarrow \sqrt{x^2 - 2x - 8} = \sqrt{(x-1)^2 - 9}$
 $(3)^2$
 $x-1 = 3 \sec t \rightarrow C$
 $x = 1 + 3 \sec t$



Aşağıda verilen belirsiz integraller ile çözümleri için yapılabilecek dönüşümler hangi seçenekte doğru olarak eşleştirilmiştir?

1. $\int \frac{dx}{2-\cos x} \rightarrow B$

a. $x = \frac{\sqrt{3}}{2} \tan t - \frac{1}{2}$

2. $\int \frac{dx}{\sqrt{x^2+x+1}} \rightarrow \sqrt{\left(x+\frac{1}{2}\right)^2 + \frac{3}{4}} \rightarrow x+\frac{1}{2} = \frac{\sqrt{3}}{2} \tan t$ b. $\tan \frac{x}{2} = t$

3. $\int \sqrt{4-3x^2} dx \rightarrow \sqrt{2^2 - (\sqrt{3}x)^2} \rightarrow \sqrt{3}x = 2 \sin t$ c. $x = 2 \sec t + 1$

4. $\int \frac{dx}{\sqrt{x^2-2x-3}} \rightarrow \sqrt{(x-1)^2 - 4} \rightarrow x-1 = 2 \sec t$ d. $x = \frac{2\sqrt{3}}{3} \sin t$

A) $\begin{matrix} 1-a \\ 2-b \\ 3-c \\ 4-d \end{matrix}$

B) $\begin{matrix} 1-b \\ 2-a \\ 3-d \\ 4-c \end{matrix}$

C) $\begin{matrix} 1-c \\ 2-b \\ 3-d \\ 4-a \end{matrix}$

D) $\begin{matrix} 1-b \\ 2-d \\ 3-c \\ 4-a \end{matrix}$

E) $\begin{matrix} 1-b \\ 2-a \\ 3-c \\ 4-d \end{matrix}$